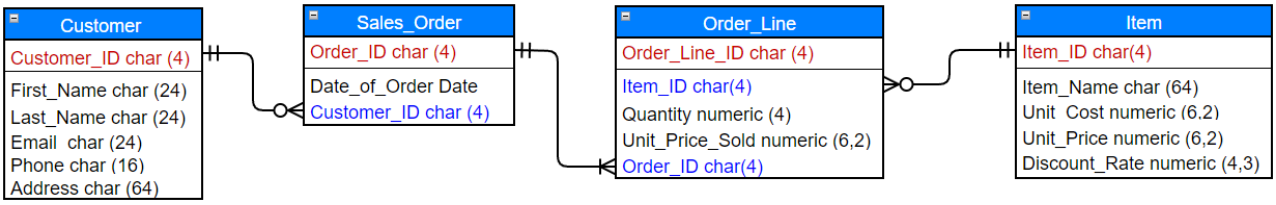
Order Management Database

The SQL Single-Table Queries lectures used the *Order Management Database* described here. You need to:

- Understand the database schema
- Use the SQL DDL statements to create the database in your database system of your choice. Note that database system vendors support a variation of the standard SQL DDL syntax. Therefore, it may be necessary to revise the given SQL DDL statements to remove errors encountered. Seek the help of your AI tool to fix the errors.

Order Management Database Schema

The Order Management Database contains four tables as shown in the diagram below.



As shown in the diagram, the database has four tables (relations). The first field (attribute) in each table is the primary key (PK) of the table and separated by a line from other fields of the table. The fields in color blue are foreign keys (FKs) referencing to the PKs of the referenced table as shown by the relationships between tables. Note each FK is named using the same name of the referenced PK for easy to understand the relationship.

The fields of the tables are explained below:

- The *Item* table stores all item data. Its PK is Item_ID. Each item has a short name (Item_Name). The unit cost (Unit_Cost) of an item is the dollar amount the company paid for the item, while the unit price (Unit_Price) is the dollar amount the company wants to sell. The discount rate (Discount_Rate) is the percentage of the unit price offered to customers if it is greater than zero. The company periodically adjusts an item's unit price and discount rate based on the market supply and demand.
- The *Customer* table keeps all customer data. Its PK is *Customer_ID*, the unique identifier of each customer. Other attributes are self-explanatory.
- The *Sales_Order* table records all customer orders. Its PK is *Order_ID*, the unique identifier of each order. The date of order (Date_Of_Order) is when an order was placed. The customer placed the order is identified by Customer_ID, the FK pointing to the PK Customer_ID in the *Customer* table, which implemented the one-to-many relationship between *Customer* and *Sales_Order* representing the business rule that a customer may have zero, one or more orders, but each order must be placed by only one customer.

- The *Order_Line* table stores the details of order lines contained in each order. Each order line has a unique identifier, the PK *Order_Line_ID*, the item sold (*Item_ID*), the number of units sold (*Quantity*), and the unit price (*Unit_Price_Sold*) which is the unit price of the item minus the discount amount if a discount was offered at the time the item was sold. *Order_ID* of an order line is the foreign key referencing the *Order_ID* of the order it belongs representing the one-to-many relationship in which an order may have one or more order lines, but an order line must belong to only one order.

Create the Order Management Database

The SQL DDL statements have been packed in a zipped file. You should download the file and unzip it on your local computer.

The steps to create the database are:

1. Run the SQL CREATE TABLE statements in the file *SQL Exercise DB Create.sql*. Do not change the order of those statements.
2. Run the SQL INSERT INTO statements in the file *SQL Exercise DB Insert.sql*, which insert test data into those tables. Do not change the order of those statements.
3. Run the SQL DELETE FROM statements in the file *SQL Exercise DB Insert.sql*. Those statements remove data from those tables. Run them only if you want to clear the data. Do not change the order of those statements.
4. Run the SQL DROP TABLE statements in the file *SQL Exercise Drop.sql*. Those statements remove all data and tables. Run them only if you want to recreate the database. Do not change the order of those statements.